

AMIT PADHAN

Email: padhanamit072006@gmail.com | Phone: +91-7505795679 | GitHub: github.com/amitpadhan07 | LinkedIn: linkedin.com/in/amitpadhan

PROFESSIONAL SUMMARY

Results-driven B.Tech Computer Science Engineering sophomore with a strong foundation in software engineering, database management, and scalable web architectures. Adept at building production-ready REST APIs, implementing secure Role-Based Access Control (RBAC), and automating cloud/local workflows. Demonstrates analytical problem-solving skills with a deep interest in Machine Learning, Generative AI, Big Data, and Cloud Computing to design intelligent, high-performance systems.

TECHNICAL SKILLS

Languages:	C++, Python, Java, JavaScript, TypeScript, HTML5, CSS3, SQL
Frontend:	React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend & Databases:	Node.js, Express.js, Next.js API Routes, MongoDB, PostgreSQL, MySQL
Tools & Platforms:	Git, GitHub, AWS, Apache, Vercel, Render, VS Code
Core Concepts:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Cloud Computing, Software Architecture, Automation, Analytics, Database Management Systems (DBMS), REST APIs, RBAC Security

ACADEMIC & PERSONAL PROJECTS

College Event Management Portal *Next.js, React, TypeScript, MongoDB, Tailwind CSS*

- **Production Architecture:** Engineered a production-ready, full-stack event portal, improving page load times, availability, and responsiveness by 35%.
- **RBAC Security:** Collaborated with stakeholders to define requirements and implement secure Role-Based Access Control (RBAC), managing permissions for admins, organizers, and attendees.
- **Database Efficiency:** Optimized MongoDB schemas and indexing, improving database query efficiency and reducing latency by 20% under peak traffic.
- **Workflow Automation:** Designed an automated registration workflow and real-time attendance analytics, decreasing check-in queue times by 50%.

PeriodTracker – Cycle Tracking & Health Analytics *Next.js, React, TypeScript, MongoDB, Recharts, Next-Auth*

- **Health Analytics:** Developed a full-stack health analytics application to monitor and track biological cycles based on user logs.
- **Data Observability:** Designed an analytics dashboard using Recharts to visualize historical health logs into trends, enhancing data observability.
- **Serverless Operations:** Created a scalable backend using Next.js API Routes and MongoDB to store, retrieve, and aggregate user records efficiently.
- **Data Security:** Implemented secure user authentication and session persistence utilizing Next-Auth to protect sensitive user profiles.

Certiit – Certificate Download Portal

React, Node.js, Express, MongoDB, PDFKit

- **PDF Automation:** Built an automated verification system that dynamically renders, encrypts, and dispatches custom PDF certificates.
- **Fraud Prevention:** Designed certificate verification lookup paths with unique SHA-256 hash identifiers to eliminate fraudulent credentials and ensure authenticity.
- **Bulk Dispatch:** Programmed admin tools supporting CSV uploads and bulk email triggers to automate the certificate dispatch workflow.

RSSB Management System

React, Node.js, Express, PostgreSQL

- **DBMS Digitization:** Designed and deployed a full-stack administrative management system to digitize operations in a cooperative team environment.
- **Process Automation:** Automated record management processes, reducing manual paperwork by 80% and reducing operational errors.
- **SQL Custom Reports:** Formulated custom SQL query templates to enable administrators to compile operational summaries on demand.

EDUCATION

B.Tech in Computer Science Engineering (AI Full Stack Focus)

Expected Graduation: 2028

Graphic Era Hill University | Dehradun, Uttarakhand

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence

CERTIFICATIONS

• NPTEL – Database Management Systems • AWS Cloud CLI Essentials • Python Programming Bootcamp • DSA & C++ Bootcamp • Git & GitHub Bootcamp • JavaScript Bootcamp • Google AI for Data Analysis • Google AI for Research and Insights • Machine Learning Workshop

KEY ACHIEVEMENTS

- **DSA Solving:** Solved 200+ data structure and algorithmic problems on LeetCode to build strong computer science fundamentals.
- **Full-Stack Deployment:** Developed and deployed 4 production-grade full-stack web applications, demonstrating SDLC mastery.
- **Workflow Automation:** Automated event workflows and record management, reducing lines and check-in wait times by 50%.